

Feasibility Analysis and Techno-Economic Study for Green Energy System Connected to Unhealthy Grid

Emad M. Sayed
Electrical Engineering Department
Faculty of Engineering-Fayoum
University-Fayoum, Egypt
ems13@fayoum.edu.eg

Khaled H. Ibrahim
Electrical Engineering Department
Faculty of Engineering-Fayoum
University-Fayoum, Egypt
khi00@fayoum.edu.eg

Ahmed Farhan
Electrical Engineering Department
Faculty of Engineering-Fayoum
University-Fayoum, Egypt
afm01@fayoum.edu.eg

Mohamed Mahmoud Samy
Dept. of Electrical Engineering
Faculty of Engineering, Beni-Suef
University, Beni-Suef, Egypt
Mohamed.samy@eng.bsu.edu.eg

Abstract— In regions plagued by unreliable grid infrastructure, optimization of hybrid renewable systems is paramount to ensure consistent power supply. This study focuses on sizing a hybrid green energy system (HGES) for a hospital in Egypt facing unhealthy utility, with a daily average load demand of 8809 kWh. Utilizing HOMER Pro software, the study simulates various configurations of photovoltaic arrays, wind turbines, battery storage, and power converters to identify an optimal setup that meets the hospital's energy requirements while mitigating the impact of unscheduled power outages. The analysis considers the hospital's specific load profile and potential grid outages, optimizing the system for maximum efficiency and reliability. The outcomes of the emulation illustrate that the optimal hybrid system achieves a noteworthy cost of energy (COE) at around \$0.0429 per kilowatt-hour. Additionally, the system displays a net present cost of \$3.76 million. These findings reveal that integrating PV and wind generation with adequate battery storage and efficient converters can significantly improve the hospital's energy independence and reduce CO2 emissions. The economic assessment within HOMER Pro indicates that the chosen configuration not only ensures a stable power supply but also achieves the lowest net present cost (NPC) and cost of energy (COE), demonstrating the practical benefits of renewable resources integrating in cases affected by frequent power interruptions.

Index Terms-- Unreliable Grid; Optimization; Green energy system (GES); COE; NPC; HOMER Pro software.

INTRODUCTION

A hospital or healthcare facility is among the commercial sector's most energy-intensive establishments, where the demand for electrical power must be consistently fulfilled around the clock, every day of the week [1]. One of the primary problems in various applications is grid power outages, which can be addressed using storage or backup systems. Hospitals have a variety of emergency procedures and energy consumption patterns, thus, to cover grid power outages, a diesel generator (DG) is employed as a support system [2]-[4]. This usage of DG results in pollution emissions, in addition, the DG frequently is unable to supply the hospital with the necessary amount of electric energy because of technical issues or a shortage of fuel. Thus, the researchers' first priority has always been the hospitals' consistent and dependable energy supply other than diesel generators.

Utilizing a hybrid green energy system (HGES) has become an indispensable and viable way to mitigate this issue and balance the load demand on hospitals [5],[6].

HOMER Pro software is widely recognized as a valuable tool for the development and assessment of hybrid energy systems. It empowers users to input parameters such as device costs, demand profiles, source information, and economic constraints to derive practical results. These results, tailored to the specific load profile, offer optimal component sizing for various resources like PV panels, wind turbines, batteries, DG generators, and converters. Furthermore, HOMER provides key metrics including NPC, energy generation, consumption figures, COE, surplus energy, and energy efficiency levels. The functionalities of this software empower decision-makers to make well-informed decisions and enhance the design of hybrid systems, leading to cost-efficient and effective energy solutions [7]-[9].

HOMER is a widely used software for optimization, especially in the field of hospital construction applications. A study of hospital positioned in a rural place of Pakistan's Azad Jammu and Kashmir focuses on the implementation of a demand calculation and economic assessment [10]. To minimize costs, identify the most efficient system configuration, and assess the effects on the environment, the study compares the current system (Grid/DG) with the proposed hybrid (PV/Wind/Battery/Converter/Grid) system using HOMER Pro. The results illustrate that the hybrid system suggested is superior in efficiency, stability, cost-effectiveness, and environmentally friendly attributes.

A research paper of rural healthcare facilities in Nigeria and South Africa delves into the utilization of renewable systems, which comprise PV panels, wind turbines, batteries, and hydrogen storage [11]. The research assessed the effectiveness of employing HOMER software for conducting techno-economic evaluations, sizing, optimizing, and selecting the optimal HGES. This study [12] uses HOMER Pro software to perform the techno-economic analysis for the proposed Model consisting of PV panels and batteries for a hospital located in Tehran, Iran. According to the study's findings, the suggested system outperformed the existing one (Grid, DG) in terms of both energy estimate and NPC.

The issue of unreliable grids is examined in this paper due to the fact that these grids often experience unscheduled power disruptions throughout the year. Using Homer Program to propose a grid-connected renewable power system consisting of PV panels, wind turbines, storage units, and converter as an alternative system employed to mitigate power failures and give green energy that is essential and preferred for the hospital environment.

METHODOLOGY

This segment delineates the process utilized to realize the research target. It elucidates the different procedures employed based on HOMER Program for modeling the suggested system. Fig. 1 clears the introduced system.

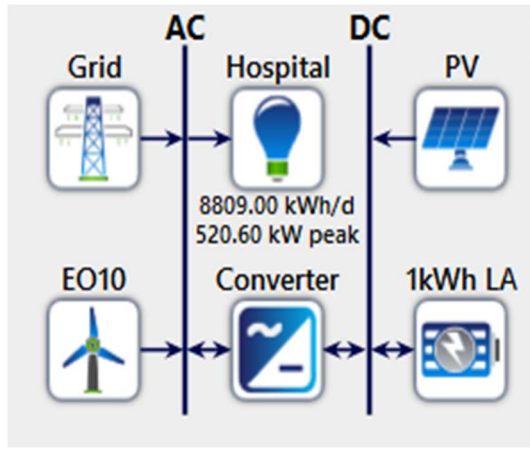


Fig. 1: Proposed system architecture.

i. Case study Description and Weather Data Inputs

The hospital under consideration is situated in Hurghada, Egypt, at a latitude of 27°15.5' N and longitude of 33°48.7' W. The Hospital requires an average daily energy consumption of approximately 8809 kWh/day, with a peak load of 520.6 kW. Fig. 2 clears the chosen position's monthly mean demand.

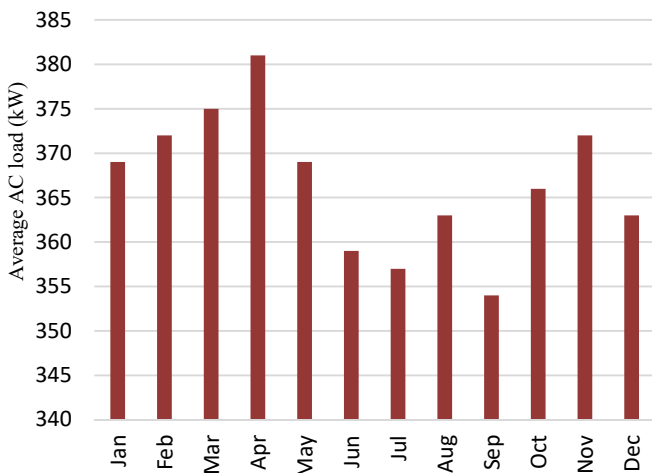


Fig. 2: Selected site's monthly average load.

The monthly data on wind speed is illustrated in Fig. 3.

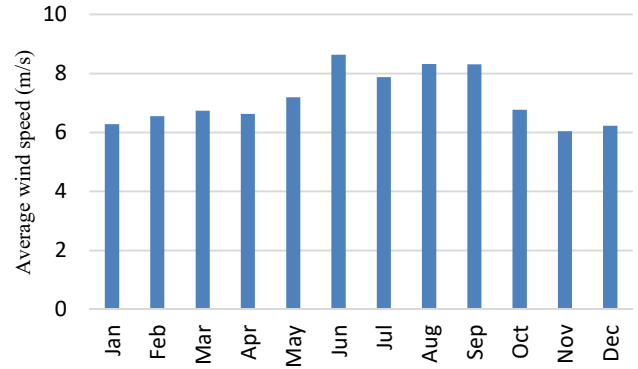


Fig. 3: The yearly mean wind speed of the selected position.

Figure 4 clears the average solar irradiance of the proposed system location with scaled annual average is 6.22 kWh/m² per day.

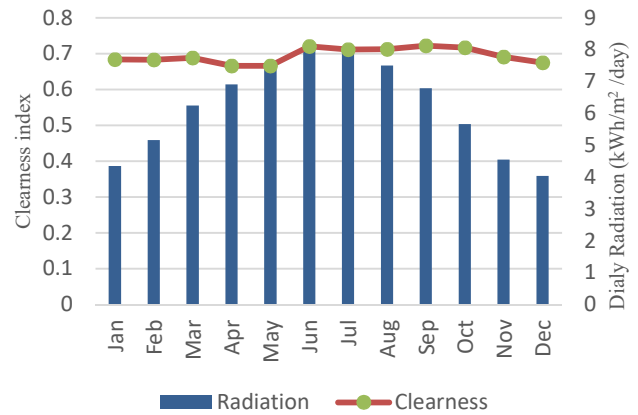


Fig. 4: The annual mean global irradiance of the selected location.

ii. Mathematical modelling of the system components.

Solar energy

HOMER is utilized to calculate the PV output power utilizing equation (1).

$$P_{PV}(t) = Y_{PV} * f_{PV} * \left(\frac{G_T(t)}{G_{T,STC}} \right) [1 + \alpha_P (T_c(t) - T_{c,STC})] \quad (1)$$

Where: Y_{PV} : the estimated PV power output at STC (kW); f_{PV} : derating parameter (%) (In this paper, the PV array derating factor is 80% [8].); G_T : irradiance (kW/m²); $G_{T,STC}$: Solar radiation at STC (1 kW/m²); α_P : Temperature parameter (% / °C); $T_{c,STC}$: Cell temperature at base conditions (25 °C); T_c : Cell temperature (°C).

The next equation can be utilized to find the temperature of the cell [10].

$$T_{cell} = \frac{T_a + (T_{c,NOCT} - T_{a,NOCT}) \left(\frac{G_T(t)}{G_{T,NOCT}} \right) \left(\frac{1 - \eta_{mp,STC} [1 - \alpha_P T_{c,STC}]}{\tau \alpha} \right)}{1 + (T_{c,NOCT} - T_{a,NOCT}) \left(\frac{G_T(t)}{G_{T,NOCT}} \right) \left(\frac{\eta_{mp,STC} \alpha_P}{\tau \alpha} \right)} \quad (2)$$

Wind turbine

The consequence power of the wind turbine (WT) system at every time interval is estimated via the HOMER program. This process encompasses a three-stage methodology, which includes.

1. Estimate the wind speed at the height of the wind turbine hub (HH) based on Eqn. (3) [11].

$$V_{hub} = V_{anem} \frac{\ln(\frac{Z_{hub}}{Z_o})}{\ln(\frac{Z_{anem}}{Z_o})} \quad (3)$$

Where V_{hub} :the wind speed at the hub height of the WT [m/s]; V_{anem} :the wind speed at anemometer height [m/s]; Z_{hub} :the hub height of the WT [m]; Z_{anem} :the anemometer height [m]; Z_o :the surface roughness length [m].

2. Evaluate the power generated by the WT at that wind speed under the standard air density. When the wind speed at HH is assessed. Figure 5 displays the power curve of the selected wind turbine.

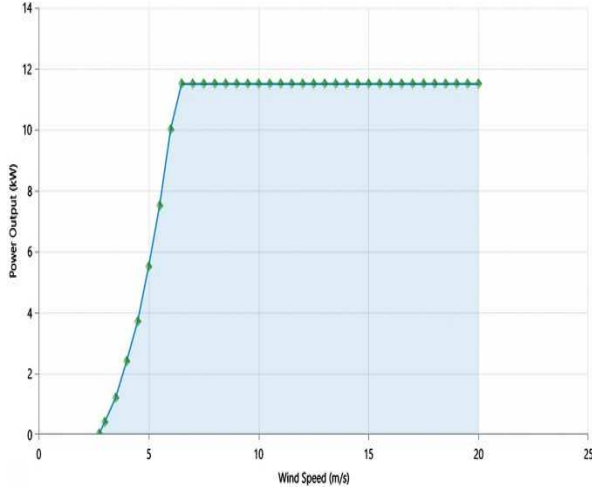


Fig. 5: Power curve of the selected WT.

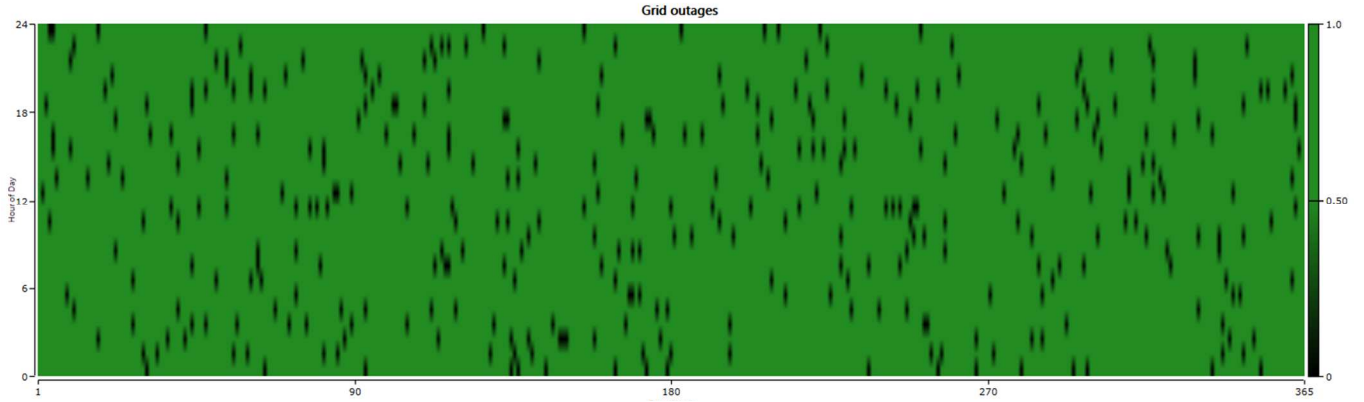


Fig. 6: daily power outages

Battery Bank

In HOMER, the lifetime of a storage bank is governed by two distinct factors: the throughput over its lifetime and the duration of its storage float. Essentially, batteries face the risk of expiration due to either operational usage or simply the passage of time. Upon developing a new storage unit, the decision lies in determining whether its lifetime is constrained by chronological age, operational throughput, or a combination of both. HOMER employs a calculation to determine the lifetime of the storage bank, which is based on the following equation (4):

$$L_{batt} = \begin{cases} \frac{N_{batt} \cdot Q_{lifetime}}{Q_{thr}} \\ \text{MIN}(\frac{N_{batt} \cdot Q_{lifetime}}{Q_{thr}}, L_{batt,f}) \end{cases} \quad (4)$$

Where L_{batt} is the yearly life time of the storage bank
 N_{batt} :batteries number.

$Q_{lifetime}$: age production of a single storage [kWh];

Q_{thr} : yearly storage production [kWh/yr];

$L_{batt,f}$: storage float age [yr].

Power inverter

The DC output from the PV systems is directed towards a DC link. The inverter is needed to connect this DC link to the AC load and the electrical grid at the AC bus. The on-grid hybrid system under consideration includes a battery as a storage component, thus necessitating the use of a bi-directional inverter. This type of inverter enables the charging of batteries from the AC side in case the output power from the hybrid system is insufficient.

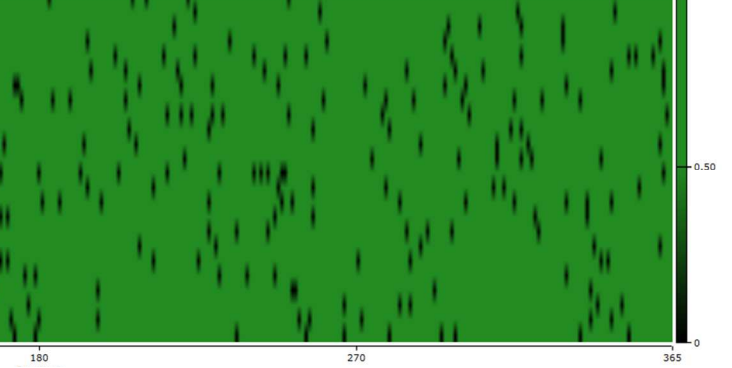
The output power (P_{out}) of the inverter can be found using equation (5):

$$P_{out} = P_{dc} * \eta_{inv} \quad (5)$$

Where η_{inv} : inverter efficiency; P_{dc} : the DC input power.

Grid Modelling

The electrical grid is considered a primary supplier of alternating current (AC) electricity to supply the load, enabling transactions of power with no limitations on buying or selling. HOMER Pro enables the modeling of an unreliable grid



through the inclusion of both scheduled and unscheduled outages. The investigation in this research focused on a model of an unreliable grid that specifically incorporated unscheduled grid outages. Fig. 6 illustrates the power outages through the year.

iii. Economical modelling of the HGES components.

In HOMER, the economic modeling revolves around mitigating the overall NPC and the COE. These fundamental economic variables are dependent on the overall estimate of the system, which is computed annually. Estimates contain the initial, replacement, O&M, fuel estimate, emissions punishments, and the estimate of the purchased power from the utility. Profits contain salvage value and utility sales revenue.

The NPC is analyzed with the help of the next Eqn..

$$NPC_{tot} = \frac{C_{ann,tot}}{CRF(r,m)} \quad (6)$$

Where $C_{ann,tot}$:is the overall yearly estimate (\$/year);

r :is the yearly benefit tariff (%);

m :is the age of the project (years), and

CRF :is the principal retrieval parameter estimated by the next Eqn.

$$CRF(r, m) = \frac{r(1+r)^m}{(1+r)^m - 1} \quad (7)$$

The yearly benefit tariff is estimated from the figurative interest and inflation tariff utilizing next Eqn.

$$r = \frac{i-f}{1-f} \quad (8)$$

Where i : is the figurative benefit tariff (%), and

f is the yearly insufflation tariff (%). In this study, a 27.25% figurative benefit tariff and a 27.5 % yearly insufflation tariff are taken according to the central bank of Egypt (CBE) [13].

The mean estimate of operative power output per kWh, is named the COE, is obtained:

$$COE = \frac{C_{ann,tot}}{E_{served}} \quad (9)$$

E_{served} is the overall demand served utilizing the next Eqn.:

$$E_{served} = E_{AC} + E_{DC} + E_{Def} + E_{Sales} \quad (10)$$

Where E_{AC} is the AC fundamental demand [kWh/yr];

E_{DC} :is the DC fundamental demand [kWh/yr];

E_{Def} :is the deferrable demand [kWh/yr],

E_{Sales} :is the power pushed to the utility [kWh/yr].

The salvage value denotes the remaining worth within a constituent of the power system upon the culmination of the project's lifetime and evaluated with the next Eqn.

$$S_{val} = R_{cost} \frac{L_{rem}}{L_{comp}} \quad (11)$$

Where S_{val} : salvage value [\$];

R_{cost} : component replacement estimate [\$];

L_{rem} : is the residual age of the device at the finish of the project lifetime[yr];

L_{comp} : device age [yr].

The economic and technical information regarding the power resources and devices integrated to the proposed system are outlined in Tables 1,2,3 and 4. These specific details are crucial for evaluating the characteristics and economic viability of the system. The tables function as a primary source for conducting thorough analyses and enhancements throughout the extended operational lifetime of the HGES, which is anticipated to be 25 years at the designated location.

For the Grid, The prevailing average electricity tariff is approximately 0.03 \$/kWh, while the sellback price is around 0.02 \$/kWh [14]. To date, the rate in the nation is supported by the authorities.

Table 1: Specifications of PV array [15]

Generic flat plate PV	
Rated capacity (KW)	1
Capital estimate (\$)	1073
O&M estimate (\$/year)	10
Replacement cost (\$)	1073
Lifetime (years)	25
Derating factor (%)	80
Efficiency (%)	13

Table 2: Wind turbine technical parameters and cost assumptions [16]

Eocycle EO10	
Rated capacity (kW)	10
Capital estimate (\$)	30000
O&M estimate (\$/year)	600
O&M estimate (\$/year)	30000
Age (years)	20
Hub height (m)	16

Table 3: Battery technical parameters and cost assumptions [15]

Generic 1kWh Lead Acid	
Nominal voltage (V)	12
Peak capacity (Ah)	83.4
Capital estimate (\$)	300
O&M estimate (\$/year)	10
Replacement cost (\$)	300
Age (years)	10
Capacity ratio (%)	40.3
Rate constant (1/hr)	0.827

Table 4: Converter technical parameters and cost assumptions [15]

Generic system converter	
Capital estimate (\$)	300
O&M estimate (\$/year)	0
Replacement cost (\$)	300
Age (years)	15
Inverter Efficiency (%)	95
Rectifier Efficiency (%)	95

OPTIMIZATION OUTCOMES AND DISCUSSION

The current paper employs the HOMER Program for conducting simulations and optimizations aimed at attaining the most efficient configurations for Hybrid-Green Energy Systems (HGES) at the selected site. The basic goal is to find the most suitable sizing for HGES characterized by a balance between cost efficiency, and the provision of a reliable power supply, through a feasibility techno-economic evaluation, considering critical parameters such as NPC and COE.

Table 5 illustrates the optimal Hybrid-Green Energy Systems (HGES) with the most cost-effective NPC, COE, and Renewable Energy (RE) fraction for the specific site. The comparative evaluation of various system configurations has been conducted considering parameters such as NPC, COE, CO2 emissions, and peak RE fraction.

From the table, it is observed that the optimal system proposed by HOMER is a well-balanced setup, consisting of 211 kW of photovoltaic panels, 32 wind turbines utilizing the high energy output due to the high wind speeds at the site, 1082 batteries each with a 1 kWh capacity, and a 388-kW converter. This particular configuration demonstrates the most cost-effective COE at \$0.042/kWh and the lowest NPC at \$3.76M. Moreover, it attains a renewable energy fraction of around 77.4%, a figure notably superior to alternative systems.

The Wind turbine-battery-converter configuration emerges as the second most economically efficient layout, identified as the optimal hybrid system with a NPC amounting to \$3.85M and a COE of \$0.044. The devices used utilized in the ideal HGES architecture are shown in Table 6 together with their generation of power, Table 7 illustrates the consumption of energy throughout the project.

Table 6: HGES power generation

Production		
Component	kWh/year	%
PV	373,907	10.8
Wind (EO10)	2,308,128	66.8
Grid	773,834	22.4
Total	3,455,870	100

Fig. 8 Clears the monthly production of various power resources utilized in HGES.

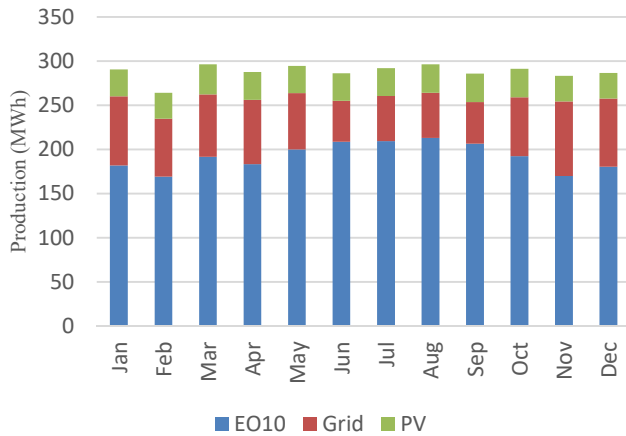


Fig. 8: The optimal HGES components monthly electric production.

Table 5: different HGE systems configuration

HGES	PV (kW)	Wind turbines (EO10)	Lead acid batteries	Converter (kW)	NPC (\$)	COE (\$)	RF (%)
PV/Wind/Batteries	211	32	1082	388	3.76M	0.042	77.4
Wind/Batteries	-	33	1208	363	3.85M	0.044	70.6
PV/Batteries	1092	-	1476	530	4.46M	0.051	49

Table 7: Power consumption

Consumption	kWh/year	%
AC Load	3,212,117	93.9
Grid Sales	206,934	6.05
Total	3,455,870	100

The cash flow of the system is illustrated in Fig. 9, encompassing capital outlays, replacement outlays, operational and maintenance (O&M) disbursements, and residual values for each constituent. It is essential to highlight the standard flat plate PV array involves an initial capital investment of \$226,517 along with yearly O&M costs amounting \$54,146.10. In contrast, the typical 10 kW wind turbine necessitates an initial capital outlay of \$960,000 and incurs annual O&M expenses totaling \$492,454.21. while the batteries' initial cost \$324,600 with an operating cost \$277,518.4. The grid infrastructure, indicative of grid-associated costs, registers a negative value of \$489,282.96, indicating potential earnings. Lastly, the converter requires an initial capital expenditure of \$116,296 and replacement expenses of \$119,771.50. with a total annualized cost \$146,606.63.

The examination exposes a slight excess of electrical power amounting to 9,123 kWh, which accounts for 0.264% of the overall output. It is important to highlight the absence of any unmet load, signifying effective management of energy supply in meeting load requirements. Likewise, there is an absence of insufficient capacity, emphasizing the system's ability to satisfy load demands.

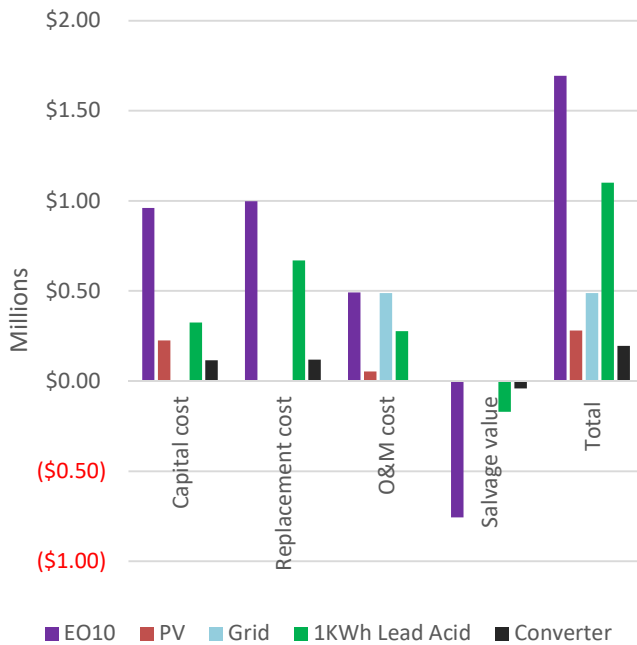


Fig. 9: The optimal HGES components cash flow.

CONCLUSIONS

This study introduces a techno-economic optimization analysis of a hybrid PV/wind renewable system designed to fulfill the electricity demand of a hospital located in Hurghada, Egypt, utilizing HOMER Program. The examination exposed that the best distribution of the PV and wind generation constitute 10.8% and 66.8% respectively of the overall energy production resulting in a reduction in CO₂ emissions of 74%. The findings of the optimization analysis indicated that the initial capital outlay totals 38.951 million dollars, with a yearly working and maintenance expenditure of 27,840 dollars, resulting in a NPC of 43.386 million dollars. Ultimately, it was decided that the energy expenditure associated with the suggested hybrid energy system comes to \$0.042 per kWh.

REFERENCES

- [1] Hussein Abubakr, Juan C Vasquez, Karar Mahmoud, Mohamed MF Darwish, Josep M Guerrero, "Comprehensive review on renewable energy sources in Egypt—current status, grid codes and future vision", IEEE Access, Vol. 10, Pp. Page(s):4081- 4101, 2022
- [2] Usman, Rilwan, and Marvin BarivureSigalo. "Analysis and Simulation of Electrical Load in a Hospital Using Hybrid (Diesel/Solar) System as a Back-Up." Journal of Electronics and Communication Engineering Research 3.9 (2017): 01-14.
- [3] Peirow, Setare, Fatemeh Razi Astaraei, and Amirali Saifoddin Asl. "Techno-economic and environmental assessment of a hybrid renewable energy system for a hospital using multi-criteria decision-making method." Energies 16.4 (2023): 1916. <https://doi.org/10.3390/en16041916>
- [4] Samy, Mohamed M., and Mohamed Nabile. "Design and Analysis of an Islanded Green Energy Station for Highways Emergency Hospitals at Different Locations in Egypt." Journal of Engineering Science and Sustainable Industrial Technology 2.2 (2024): 17-27. [10.21608/jessit.2023.224345.1001](https://doi.org/10.21608/jessit.2023.224345.1001)
- [5] Ansari, Amir Babak. "Multi-objective size optimization and economic analysis of a hydrogen-based standalone hybrid energy system for a health care center." International Journal of Hydrogen Energy 62 (2024): 1154-1170. <https://doi.org/10.1016/j.ijhydene.2024.03.165>
- [6] Naveed, Ausnain, et al. "Renewable energy integration in healthcare systems: A case study of a hospital in Azad Jammu and Kashmir." IET Renewable Power Generation 18.5 (2024): 796-809. <https://doi.org/10.1049/rpg2.12946>
- [7] Youssef, Amr A., et al. "Islanded green energy system optimal analysis

using PV, wind, biomass, and battery resources with various economic criteria." Results in Engineering 19 (2023): 101321. <https://doi.org/10.1016/j.rineng.2023.101321>

- [8] Mousa, Mostafa M., et al. "Techno-Economic Analysis and Simulation of Electric Vehicle Charging Stations based on Green Energy System." 2023 24th International Middle East Power System Conference (MEPCON). IEEE, 2023. [10.1109/MEPCON58725.2023.10462477](https://doi.org/10.1109/MEPCON58725.2023.10462477)
- [9] Rice, Ilunga Kajila, et al. "A Hybrid Photovoltaic/Diesel System for Off-Grid Applications in Lubumbashi, DR Congo: A HOMER Pro Modeling and Optimization Study." Sustainability 15.10 (2023): 8162. <https://doi.org/10.3390/su15108162>
- [10] Naveed, Ausnain, et al. "Renewable energy integration in healthcare systems: A case study of a hospital in Azad Jammu and Kashmir." IET Renewable Power Generation 18.5 (2024): 796-809. <https://doi.org/10.1049/rpg2.12946>
- [11] Adedoja, Oluwaseye Samson, Emmanuel Rotimi Sadiku, and Yskandar Hamam. "A techno-economic assessment of the viability of a photovoltaic-wind-battery storage-hydrogen energy system for electrifying primary healthcare centre in Sub-Saharan Africa." Energy Conversion and Management: X (2024): 100643. <https://doi.org/10.1016/j.ecmx.2024.100643>
- [12] Salau, Ayodeji Olalekan, et al. "Design and Analysis of a Sustainable Power System for Healthcare Centers Using Renewable Energy Sources." Zhongguo Shiyu Daxue Xuebao (Ziran Kexue Ban)/Journal of China University of Petroleum (Edition of Natural Science) 47.4 (2023): 1249-1260.
- [13] Central Bank of Egypt (CBE), Online.Available <http://www.cbe.org.eg/en/Pages/default.aspx>.
- [14] Egyptian Electric Utility for Consumer Protection and Regulatory Agency, Renewable Energy Feed-in Tariff Projects' Regulations online, available at: <http://egyptera.org/en/t3reefa.aspx>.
- [15] Riayatsyah, T. M. I., et al. "Techno-economic analysis of hybrid diesel generators and renewable energy for a remote island in the indian ocean using HOMER Pro." Sustainability 14.16 (2022): 9846. <https://doi.org/10.3390/su14169846>
- [16] Duman, A. Can, and Önder Güler. "Techno-economic analysis of off-grid PV/wind/fuel cell hybrid system combinations with a comparison of regularly and seasonally occupied households." Sustainable Cities and Society 42 (2018): 107-126. <https://doi.org/10.1016/j.scs.2018.06.029>